2.5.1 Quadratic Formula

Is it possible to write the general quadratic function $P(x) = ax^2+bx+c$ in the form $a(x-k)^2+d$? The answer is yes. We can do this by the method called "**completing the square.**" We shall rewrite the function $P(x)$ as follows:

$$\begin{aligned} P(x) &= ax^2+bx+c \\ &= a\left(x^2+2x\frac{b}{2a}+\frac{c}{a}\right) \\ &= a\left(x^2+2x\frac{b}{2a}+\left(\frac{b}{2a}\right)^2-\frac{b^2}{4a^2}+\frac{c}{a}\right) \\ &= a\left(x+\frac{b}{2a}\right)^2-a\frac{b^2}{4a^2}+c \\ &= a\left(x+\frac{b}{2a}\right)^2+\left(a\left(\frac{b}{2a}\right)^2-b\frac{b}{2a}+c\right). \end{aligned}$$

$$\text{Thus,} \quad P(x) = a\left(x+\frac{b}{2a}\right)^2+P\left(\frac{-b}{2a}\right). \qquad (1)$$

Now, to find the x- intercepts of the curve described by $P(x)$, let us solve for $P(x)=0$.

Considering $P(x)=0$ from (1) it follows that $a\left(x+\frac{b}{2a}\right)^2+P\left(\frac{-b}{2a}\right)=0$.

$$\begin{aligned} a\left(x+\tfrac{b}{2a}\right)^2 &= -P\left(\tfrac{-b}{2a}\right) \\ &= -\tfrac{(-b^2+4ac)}{4a} \\ \left(x+\tfrac{b}{2a}\right)^2 &= \tfrac{b^2-4ac}{4a^2}. \end{aligned}$$

So $x=\frac{\sqrt{b^2-4ac}}{2a}-\frac{b}{2a}$ or $x=-\frac{\sqrt{b^2-4ac}}{2a}-\frac{b}{2a}$.

Hence, $x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$; which is called the ***quadratic formula***.

Remark:

(i) Note that $\sqrt{u}$ is defined as a real number only for $u \geq 0$.
(ii) when we write $\sqrt{u}$, we mean only the nonnegative root.

Note that $P(x)=0$ has two distinct real solutions if $b^2-4ac>0$, the roots are real and equal if $b^2-4ac=0$, and no real root if $b^2-4ac<0$.
Thus the curve intersects x-axis in two places if $b^2-4ac>0$, touches x-axis at only one point if $b^2-4ac=0$, and does not intersect x-axis at any point if $b^2-4ac<0$.
That is why $D=b^2-4ac$ is called the ***discriminant*** of the quadratic function $P(x)=ax^2+bx+c$.

(i) If α and β are roots of $ax^2+bx+c=0$, then $\alpha+\beta=\frac{-b}{a}$ and $\alpha\beta=\frac{c}{a}$.
(ii) If the discriminant b^2-4ac is negative, then the quadratic equation $ax^2+bx+c=0$, has no real roots. In this case, we have complex roots given by

$$\alpha=\frac{-b+i\sqrt{4ac-b^2}}{2a}, \quad \beta=\frac{-b-i\sqrt{4ac-b^2}}{2a}, \quad \text{where } i^2=-1,$$

which we will study in Higher Secondary Second year
(iii) For example, let us look at the graph of $y=x^2-4x+5$. (See Figure 2.4.) Since the graph does not intersect the $x-$axis, $x^2-4x+5=0$ has no real roots.
(iv) We have the following table describing the nature of the roots of a quadratic equation and the sign of the discriminant $D=b^2-4ac$.